MLOps Assignment 2

LLMOps: API Cost, Latency, and Prompt Version Management

Name	Meraj Hassan
BITS ID	2025ae05085
BITS Mail ID	2025ae05085@wilp.bits-pilani.ac.in

Individual experiential learning assignment

Executed model: Qwen/Qwen2.5-0.5B-Instruct

Report prose: 546 words

1. Approach

EduAssist needs an academic question-answering service whose prompt changes can be measured and reproduced. I evaluated the three required templates on the same five questions. The executed backend was Qwen2.5-0.5B-Instruct through Transformers because no remote API credential was configured. The notebook also includes an optional API client that reads its key from an environment variable. Generation was deterministic and limited to 140 new tokens. MLflow stored each prompt version, its commit message, the model configuration, and one evaluation run per version.

2. Latency results

Prompt v3 was used for the five-request latency table because it is the production candidate. Mean latency was 3.383 seconds, the median was 2.563, and the observed range was 1.247 to 5.672 seconds. Sample standard deviation was 1.978 seconds. The slowest v3 request was Q1 at 5.672 seconds with 100 output tokens. Autoregressive decoding produces output sequentially, so longer answers usually take more time. CPU scheduling and memory bandwidth can also change request latency. Model loading was excluded because a persistent service loads weights before accepting traffic.

Mean	Median	Minimum	Maximum	Std. deviation
3.383s	2.563s	1.247s	5.672s	1.978s

3. Token use and cost

The selected local model has no token-based API charge. I estimated compute cost from measured generation time and the Hugging Face CPU Upgrade rate of USD 0.03 per hour, accessed on 1 November 2026. Prompt v3 averaged 79.0 input tokens and 62.0 output tokens. Its estimated mean request cost was USD 0.00002819, which projects to USD 0.2819 for 10,000 requests. Input and output cost shares were allocated by token count for the requested table. Reducing the output limit for short definitions would lower latency and local compute cost.

4. Prompt comparison

The MLflow experiment contains three runs named prompt_v1, prompt_v2, and prompt_v3. Versions 1 and 2 produced prose, so their strict JSON rates were zero. Version 3 reached a valid JSON rate of 100% after applying a decoder stop condition at the first complete JSON object. Its keyword content-check rate was 80%. This lightweight check identifies missing topic terms, but it cannot establish academic correctness. The screenshots record the run table, v3 metrics, registry history, side-by-side prompt comparison, and production alias.

Version	Latency (s)	Input tokens	Output tokens	Cost (USD)	Valid JSON	Recommended
v1	7.880	23.0	140.0	0.00006567	0%	No
v2	7.269	45.0	140.0	0.00006058	0%	No
v3	3.383	79.0	62.0	0.00002819	100%	Yes

5. Promotion decision

Prompt v3 was promoted because the application requires output that downstream code can parse without repair. A response passed only when JSON parsing succeeded, all three required fields were present, confidence used an allowed value, and key_points was a list of strings. Version 3 also had lower measured average latency and lower estimated cost than the two prose versions in this run because the JSON completion control stopped generation as soon as the object closed. Its longer input prompt therefore did not cause a higher total cost. The production alias points to registry version 3 and can be reassigned without changing application code. Semantic quality remains the main limitation. One response passed the keyword check while giving an incomplete medical-screening explanation, so a human-reviewed answer set should be added before a real deployment. I would keep the JSON release condition, add correctness scoring for every question, and reject versions that fall below either threshold.

6. Reproducibility and scope

The project stores the executed notebook, exported HTML, two required CSV files, the MLflow database and artifacts, screenshots, pinned requirements, and reproduction instructions. No secret or billing record is included. The local cost estimate excludes idle time, storage, data transfer, and tax, so it should be treated as a controlled

comparison rather than an invoice forecast.

References

MLflow Prompt Registry documentation, https://mlflow.org/docs/latest/genai/prompt-registry/

Qwen2.5-0.5B-Instruct model card, https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct

Hugging Face hardware pricing, https://huggingface.co/docs/hub/spaces-gpus

7. MLflow evidence

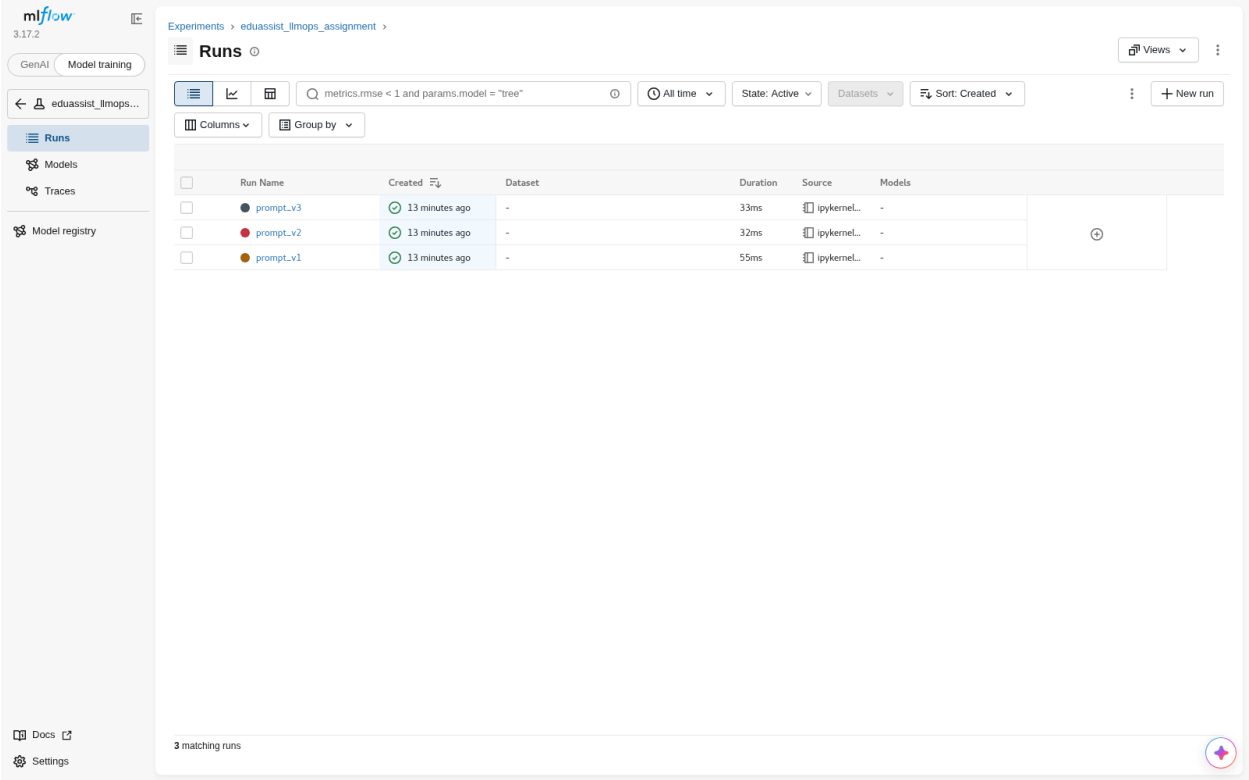


Figure 1. The MLflow experiment contains the three required prompt evaluation runs.

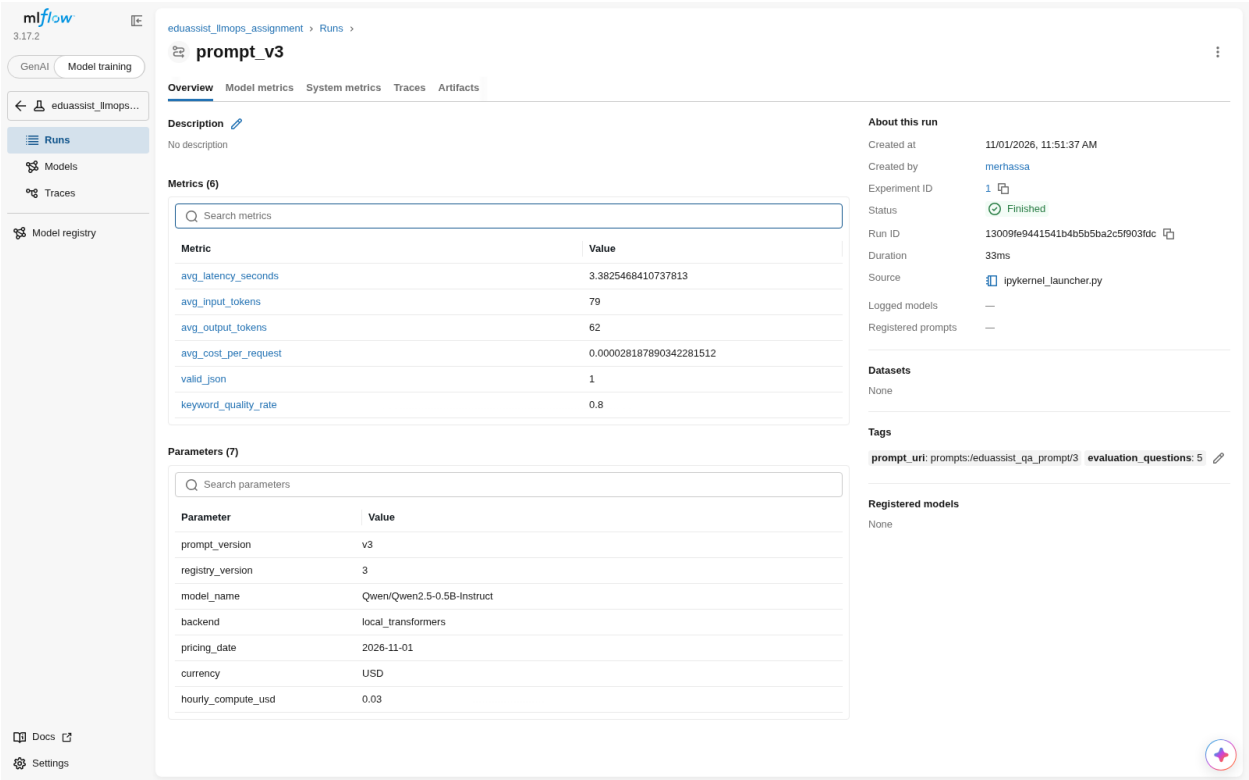


Figure 2. Logged metrics and parameters for prompt v3.

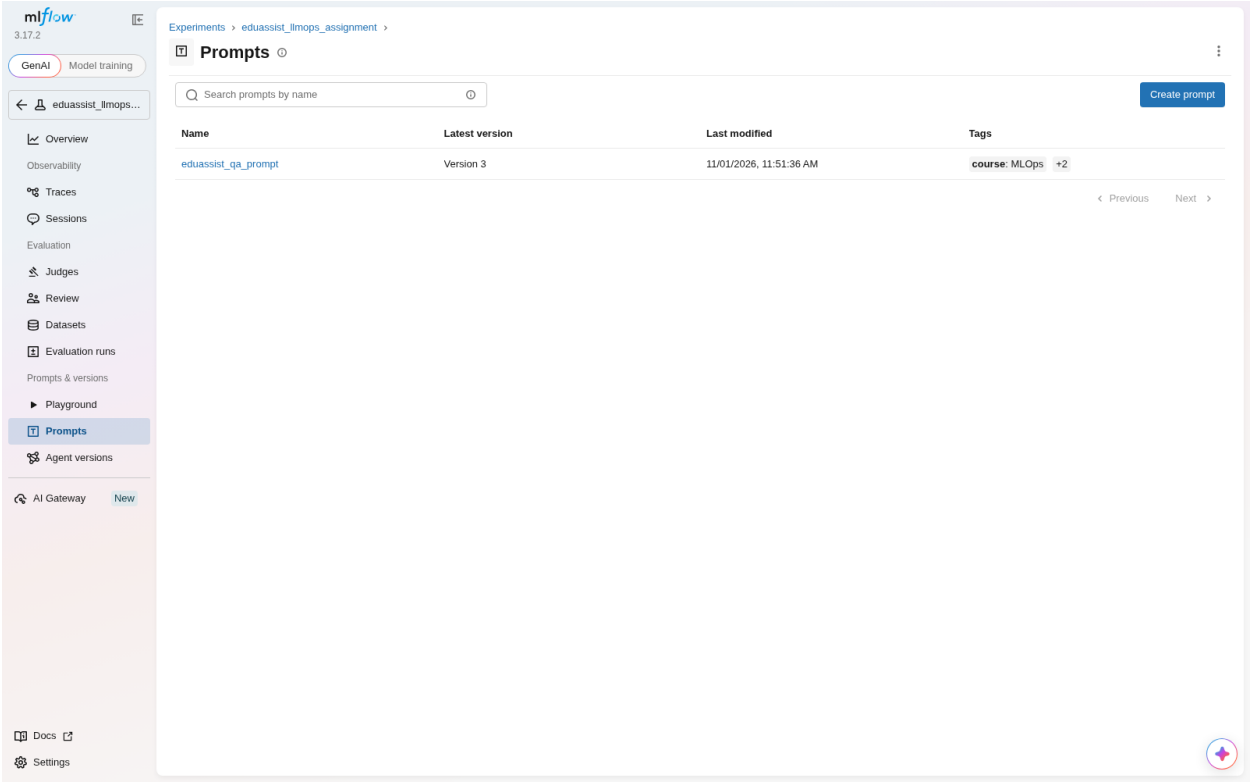


Figure 3. Prompt Registry entry for eduassist_qa_prompt.

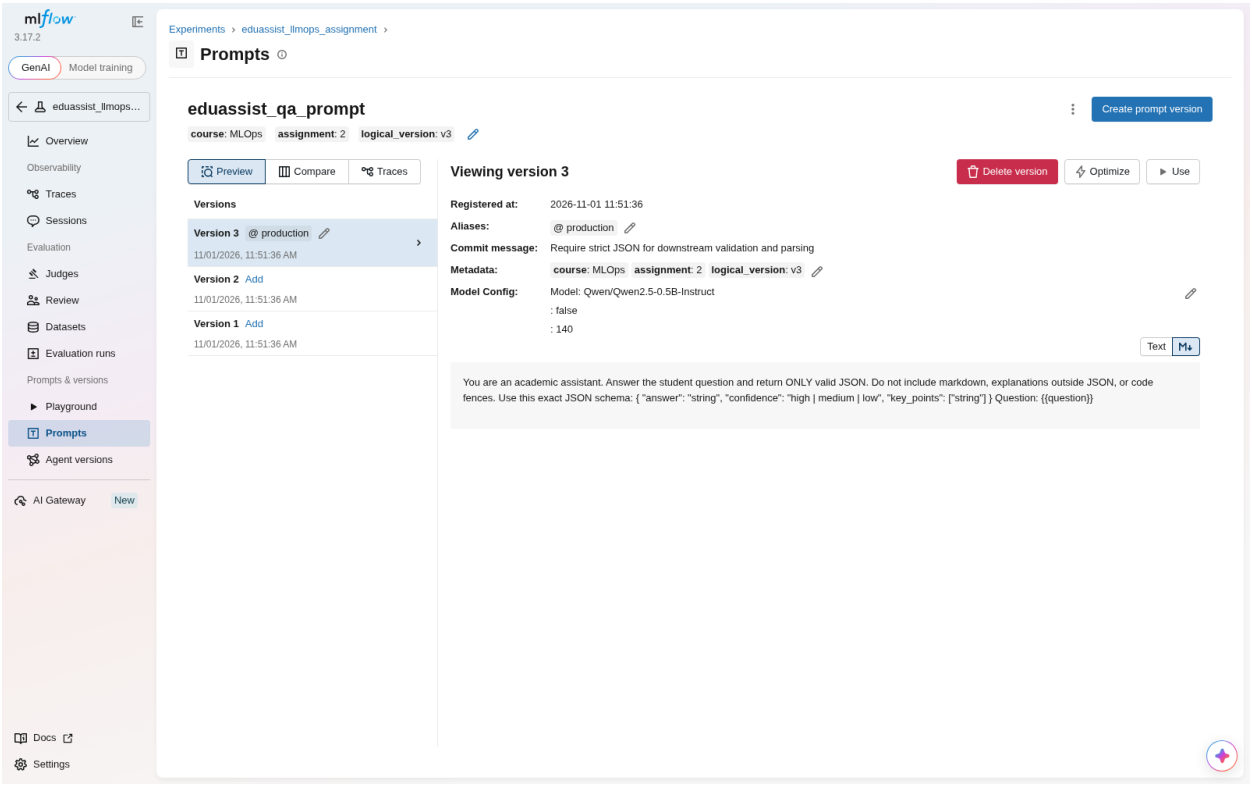


Figure 4. Versions 1, 2, and 3, with production assigned to version 3.

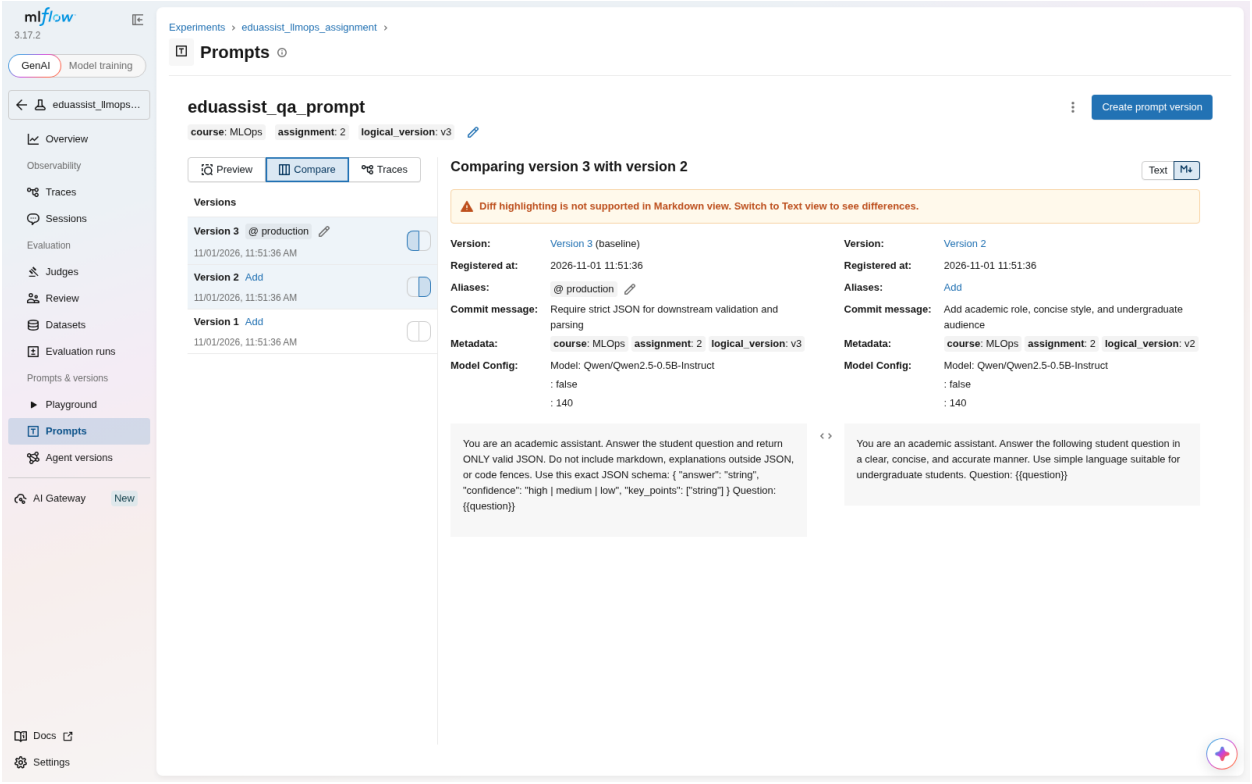


Figure 5. Side-by-side comparison of prompt versions 3 and 2.